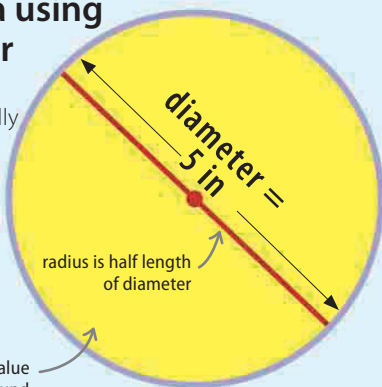


Finding area using the diameter

The formula for the area of a circle usually uses the radius, but the area can also be found if the diameter is given.



the area is the value that needs to be found

The formula for the area of a circle is always the same, whatever values are known.

$$\text{area} = \pi r^2$$

Substitute the known values into the formula—the radius is half the diameter—2.5 in this example.

$$\text{area} = 3.14 \times 2.5^2$$

the radius is half the diameter: $5 \div 2 = 2.5$

π is 3.14 to 3 significant figures

$$\text{area} = 3.14 \times 6.25$$

$2.5 \times 2.5 = 6.25$

Multiply the radius by itself (square it) as shown by the formula—this makes the last multiplication simpler.

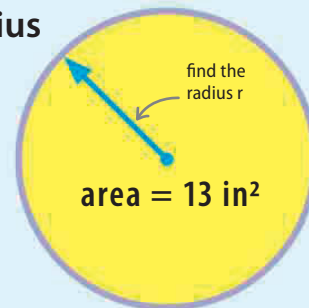
Make sure the answer is in the right units, in² here, and round it to a suitable number.

$$\text{area} = 19.63 \text{ in}^2$$

19.6349... is rounded to 2 decimal places

Finding the radius from the area

The formula for area of a circle can also be used to find the radius of a circle if its area is given.



The formula for the area of a circle can be used to find the radius if the area is known.

$$\text{area} = \pi r^2$$

Substitute the known values into the formula—here the area is 13 in².

$$13 = 3.14 \times r^2$$

divide this side by 3.14

r^2 was multiplied by 3.14, so divide by 3.14 to isolate r^2

$$\frac{13}{3.14} = r^2$$

Rearrange the formula so r^2 is on its own on one side: divide both sides by 3.14.

Round the answer, and switch the sides so that the unknown, r^2 , is shown first.

$$r^2 = 4.14$$

4.1380... is rounded to 2 decimal places

$$\sqrt{r^2} = \sqrt{4.14}$$

Find the square root of the last answer in order to find the value of the radius.

Make sure the answer is in the right units (in here) and round it to a suitable number.

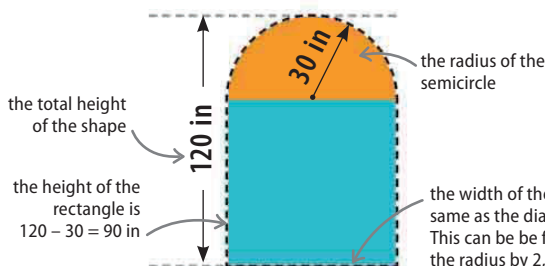
2.0342... is rounded to 2 decimal places

$$r = 2.03 \text{ in}$$

LOOKING CLOSER

Areas of compound shapes

When two or more different shapes are put together, the result is called a compound shape. The area of a compound shape can be found by adding the areas of the parts of the shape. In this example, the two different parts are a semicircle, and a rectangle. The total area is 1,414 in² (area of the semicircle, which is $\frac{1}{2} \times \pi r^2$, half the area of a circle) + 5,400 in² (the area of the rectangle) = 6,814 in².



Compound shapes

This compound shape consists of a semicircle and a rectangle. Its area can be found using only the two measurements given here.